Companion XIII

The Neutrino Sector in the Relaxation-Driven Cyclic Cosmology

Mass Hierarchy, Dirac/Majorana Structure, and Relaxation-Induced Mass Suppression

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Abstract

We develop the neutrino sector of the Relaxation-Driven Cyclic Cosmology (RDCC), extending the bipartite CPT-symmetric global state to include a dynamically generated neutrino mass matrix. While RDCC predicts a mild excess in the effective number of relativistic species, $\Delta N_{\mathrm{eff}} \sim 0.1$–$0.3$, the framework has so far remained agnostic regarding the absolute neutrino mass scale, the mass hierarchy, and the Dirac/Majorana nature of neutrinos.

In this Companion we show that the inter-sectoral relaxation mechanism introduced in Companion X naturally generates a suppressed neutrino mass scale through a universal relaxation-induced seesaw. The bipartite structure of the global state leads to a sectoral mass splitting, while the emergent thermodynamic arrows imply distinct effective masses in the two sectors. We derive the resulting mass matrix, identify the conditions for normal and inverted hierarchy, and show that the Dirac/Majorana ambiguity becomes an emergent property of the projection onto a single thermodynamic arrow.

The framework yields concrete predictions for cosmology and laboratory experiments: correlations between ΔN_{eff} and the lightest neutrino mass, a lower bound on the sum of neutrino masses, suppressed rates for neutrinoless double beta decay, and characteristic signatures in oscillation experiments. These results close the second major open gap in the RDCC program and establish the neutrino sector as a natural probe of relaxation-driven cosmology.

1 Motivation

Neutrinos occupy a unique position in modern cosmology and particle physics. They are the only Standard Model particles whose absolute mass scale remains unknown, the only fermions that may be either Dirac or Majorana in nature, and the only sector where the mass hierarchy is not experimentally established. At the same time, neutrinos constitute the second most abundant form of matter in the Universe and play a central role in structure formation, cosmic microwave background physics, and the thermal history of the early Universe.

Within the Relaxation-Driven Cyclic Cosmology (RDCC), neutrinos acquire an even more prominent role. The framework predicts a mild excess in the effective number of relativistic species, $\Delta N_{\mathrm{eff}} \sim 0.1$–$0.3$, arising from the bipartite global state and the temperature asymmetry between the two sectors. This prediction is robust and already connects RDCC to precision cosmology. However, the deeper structure of the neutrino sector has remained an open question in the RDCC program.

The central motivation of this Companion is to close this gap. The relaxation mechanism introduced in Companion X provides a natural pathway to generate suppressed mass scales without introducing new fields or explicit seesaw structures. When applied to neutrinos, the intersectoral relaxation leads to a universal suppression of effective masses, a sectoral mass splitting, and an emergent Dirac/Majorana ambiguity tied to the projection onto a single thermodynamic arrow.

Developing this structure is essential for three reasons:

- **Cosmological consistency:** The neutrino mass scale and hierarchy influence structure formation, the CMB damping tail, and the interpretation of $\Delta N_{\rm eff}$ within RDCC.
- **Theoretical completeness:** The bipartite global state naturally accommodates neutrino mixing between sectors, but the resulting mass matrix has not yet been derived.
- **Observational predictions:** Neutrinoless double beta decay, oscillation experiments, and cosmological mass bounds provide direct tests of the relaxation-induced neutrino sector.

This Companion establishes the neutrino sector as a key probe of relaxation-driven cosmology and provides the missing theoretical foundation for interpreting neutrino-related observables within the RDCC framework.

2 The RDCC Framework and the Role of Neutrinos

The Relaxation-Driven Cyclic Cosmology (RDCC) is built upon a bipartite, CPT-symmetric global state in which both sectors evolve with opposite thermodynamic arrows. The observable Universe corresponds to a projection onto one of these arrows, while the CPT-conjugate sector evolves with decreasing entropy. This structure ensures global CPT invariance while allowing for an emergent arrow of time within each sector.

The relaxation mechanism introduced in Companion X plays a central role in shaping the thermal and dynamical history of both sectors. It governs the transfer of energy, the freeze-out of sector-specific quantities, and the emergence of effective parameters such as the temperature ratio between the sectors. This mechanism also provides a natural pathway for generating suppressed mass scales without invoking new fields or explicit high-scale seesaw structures.

Neutrinos are uniquely sensitive to these features of RDCC for several reasons:

- **Thermal sensitivity:** Neutrinos decouple early and retain memory of the thermal history of both sectors. The predicted temperature asymmetry directly affects their contribution to $\Delta N_{\rm eff}$.
- **Mass generation:** The relaxation mechanism naturally induces suppressed mass scales. Neutrinos, being the lightest fermions, are the most responsive to such suppression effects.
- **Bipartite mixing:** The global state allows for mixing between the two sectors at the level of the neutrino mass matrix. This mixing leads to sectoral mass splittings and an emergent Dirac/Majorana ambiguity.
- **Observational leverage:** Neutrino masses influence structure formation, the CMB damping tail, and laboratory observables such as neutrinoless double beta decay. This makes the neutrino sector an ideal testing ground for RDCC predictions.

In this section we summarize the aspects of RDCC most relevant for neutrino physics and prepare the ground for deriving the relaxation-induced neutrino mass matrix. The key ingredients are the bipartite global state, the temperature asymmetry between sectors, and the universal relaxation dynamics that govern the evolution of both sectors across the cycle.

3 Bipartite Structure and Neutrino Degrees of Freedom

The RDCC framework is built upon a bipartite global state,

$$|\Psi_{\text{global}}\rangle = |\Psi_{\text{A}}\rangle \otimes |\Psi_{\text{B}}\rangle\,, \tag{1}$$

where the two sectors A and B evolve with opposite thermodynamic arrows. The observable Universe corresponds to a projection onto sector A, while sector B evolves with decreasing entropy. The global state is CPT-symmetric, and the two sectors are related by a CPT transformation,

$$|\Psi_{\text{B}}\rangle = CPT\,|\Psi_{\text{A}}\rangle\,. \tag{2}$$

This structure has profound implications for neutrinos. Unlike charged fermions, neutrinos are electrically neutral and therefore admit both Dirac and Majorana mass terms. In a bipartite setting, this neutrality allows for mixing between the two sectors at the level of the mass matrix. The global state therefore contains two copies of each neutrino degree of freedom,

$$\nu_{\text{A}}, \qquad \nu_{\text{B}}, \tag{3}$$

which are related by CPT but evolve in thermodynamically distinct environments.

The relaxation mechanism introduced in Companion X provides a natural coupling between the sectors. While this coupling is extremely weak for most fields, it is most effective for neutrinos due to their small masses and early decoupling. As a result, the neutrino sector becomes the primary channel through which the bipartite structure leaves observable imprints.

Three features of the bipartite state are particularly relevant:

- **Sectoral duplication:** Each neutrino species exists in both sectors, leading to a 2×2 structure in the mass matrix even before flavor mixing is included.
- **Opposite thermodynamic arrows:** The effective masses in the two sectors differ due to the distinct thermal histories. This leads to a sectoral mass splitting that is absent in conventional cosmology.
- **Relaxation-induced mixing:** The inter-sectoral relaxation introduces off-diagonal terms in the mass matrix. These terms are suppressed by the relaxation rate and naturally generate small effective masses in the observable sector.

The combination of these features leads to a universal relaxation-induced seesaw mechanism. Unlike traditional seesaw models, no new heavy fields are required; the suppression arises from the structure of the global state and the dynamics of relaxation. In the next section we derive the resulting mass matrix and identify the conditions under which the hierarchy and the Dirac/Majorana nature of neutrinos emerge.

4 Relaxation-Induced Neutrino Mass Matrix

The bipartite structure of the global state implies that each neutrino species appears in both sectors, ν_{A} and ν_{B}. In the absence of inter-sectoral interactions, the most general mass matrix would be block diagonal. However, the relaxation mechanism introduced in Companion X induces a small but non-vanishing coupling between the sectors. This leads to a 2×2 mass matrix of the form

$$M_\nu = \begin{pmatrix} m_{\text{A}} & \varepsilon \\ \varepsilon & m_{\text{B}} \end{pmatrix}, \tag{4}$$

where m_{A} and m_{B} denote the effective masses in the two sectors and ε encodes the relaxation-induced mixing.

The diagonal entries differ because the two sectors evolve with opposite thermodynamic arrows and therefore experience distinct thermal histories. The temperature asymmetry between the sectors implies

$$m_{\mathrm{B}} = m_{\mathrm{A}}\, f(T_{\mathrm{B}}/T_{\mathrm{A}}), \tag{5}$$

where f is a smooth function determined by the relaxation dynamics. For the temperature ratio predicted by RDCC, $T_{\mathrm{B}}/T_{\mathrm{A}} \simeq 0.5$, the function f typically yields a mass splitting of order unity.

The off-diagonal term ε arises from the inter-sectoral relaxation rate Γ_{rel} of Companion X. Since the relaxation mechanism is universal and extremely weak at late times, the mixing term is naturally suppressed,

$$\varepsilon \propto \Gamma_{\mathrm{rel}} \ll m_{\mathrm{A}}, m_{\mathrm{B}}. \tag{6}$$

This suppression leads to a relaxation-induced seesaw mechanism: the lightest eigenvalue of the matrix (4) is given by

$$m_{\mathrm{light}} \simeq m_{\mathrm{A}} - \frac{\varepsilon^2}{m_{\mathrm{B}} - m_{\mathrm{A}}}, \tag{7}$$

which is naturally small even if the diagonal entries are of order $10^{-1}\,\mathrm{eV}$. No heavy fields or high-scale physics are required; the suppression arises purely from the structure of the global state and the relaxation dynamics.

The heavier eigenvalue is

$$m_{\mathrm{heavy}} \simeq m_{\mathrm{B}} + \frac{\varepsilon^2}{m_{\mathrm{B}} - m_{\mathrm{A}}}, \tag{8}$$

which remains close to the sector-B mass scale. This structure implies a sectoral hierarchy: the observable sector inherits the lighter eigenstate, while the CPT-conjugate sector contains the heavier one.

The mass matrix (4) also determines the Dirac/Majorana nature of neutrinos. If the off-diagonal term dominates over the difference between the diagonal entries, the eigenstates behave effectively as Majorana fermions. If the diagonal entries dominate, the eigenstates behave as Dirac fermions. In RDCC, the relaxation-induced mixing is small but non-zero, placing the system in an intermediate regime where the Dirac/Majorana distinction becomes an emergent property of the projection onto a single thermodynamic arrow.

This relaxation-induced mass matrix forms the basis for the hierarchy analysis and the observational predictions developed in the following sections.

5 Mass Hierarchy in RDCC: Normal vs. Inverted

The relaxation-induced mass matrix derived in the previous section naturally leads to a sectoral hierarchy between the two eigenstates. The lighter eigenvalue is predominantly associated with sector A, while the heavier one is associated with sector B. This structure provides a natural starting point for understanding the observed neutrino mass hierarchy.

To connect the sectoral hierarchy to the flavor hierarchy, we extend the 2×2 mass matrix of Eq. (4) to include flavor mixing. For each flavor index $i = 1, 2, 3$, the mass matrix takes the form

$$M_{\nu,i} = \begin{pmatrix} m_{\mathrm{A},i} & \varepsilon_i \\ \varepsilon_i & m_{\mathrm{B},i} \end{pmatrix}, \tag{9}$$

where the diagonal entries differ due to the temperature asymmetry between the sectors and the off-diagonal entries are suppressed by the relaxation rate.

The eigenvalues are

$$m_{i,\pm} = \frac{1}{2}\left[m_{\mathrm{A},i} + m_{\mathrm{B},i} \pm \sqrt{(m_{\mathrm{B},i} - m_{\mathrm{A},i})^2 + 4\varepsilon_i^2} \right]. \tag{10}$$

Since ε_i is small, the splitting is dominated by the difference between $m_{\mathrm{A},i}$ and $m_{\mathrm{B},i}$. The lighter eigenvalue $m_{i,-}$ is associated with sector A and the heavier one $m_{i,+}$ with sector B.

The observed mass hierarchy depends on how the three flavor indices map onto the sectoral eigenstates. Two possibilities arise:

- **Normal hierarchy (NH):**

$$m_{1,-} < m_{2,-} < m_{3,-},$$

  corresponding to the lightest state being predominantly associated with the electron flavor.

- **Inverted hierarchy (IH):**

$$m_{3,-} < m_{1,-} < m_{2,-},$$

  corresponding to the lightest state being predominantly associated with the third flavor.

In RDCC, the hierarchy is determined by the ordering of the diagonal entries $m_{\mathrm{A},i}$. Since these entries depend on the thermal history of sector A, the hierarchy is ultimately controlled by the relaxation dynamics. The temperature asymmetry between the sectors implies that

$$m_{\mathrm{B},i} > m_{\mathrm{A},i}, \tag{11}$$

for all flavors i, but it does not fix the ordering of the $m_{\mathrm{A},i}$ among themselves.

However, the relaxation mechanism tends to suppress masses more strongly for lighter flavors, because the relaxation rate scales inversely with the mass scale. This leads to the generic expectation

$$m_{\mathrm{A},1} < m_{\mathrm{A},2} < m_{\mathrm{A},3}, \tag{12}$$

which corresponds to the normal hierarchy. The inverted hierarchy requires a fine-tuned cancellation between the diagonal entries and the off-diagonal mixing terms, making it less natural within RDCC.

Thus, while both hierarchies are in principle possible, the relaxation-induced structure of the mass matrix favors the normal hierarchy. This preference is reflected in the predictions for the sum of neutrino masses, the effective mass in neutrinoless double beta decay, and the scale dependence of neutrino free-streaming, which we discuss in the following sections.

6 Dirac vs. Majorana as an Emergent Property

In conventional particle physics, the question of whether neutrinos are Dirac or Majorana fermions is a fundamental one. Dirac neutrinos require distinct left- and right-handed fields, while Majorana neutrinos are their own antiparticles. The distinction is encoded directly in the structure of the mass terms in the Lagrangian.

In the RDCC framework, this distinction takes on a different character. The bipartite global state contains two copies of each neutrino degree of freedom, ν_{A} and ν_{B}, which are related by a CPT transformation. The observable Universe corresponds to a projection onto sector A, while the CPT-conjugate sector B evolves with the opposite thermodynamic arrow. As a result, the Dirac/Majorana nature of neutrinos becomes an emergent property of this projection rather than a fundamental feature of the underlying theory.

The relaxation-induced mass matrix of Eq. (4) contains both diagonal and off-diagonal terms. The diagonal terms correspond to sector-specific mass contributions, while the off-diagonal terms encode the inter-sectoral mixing induced by the relaxation dynamics. The relative size of these terms determines the effective nature of the neutrino eigenstates.

If the off-diagonal mixing dominates over the difference between the diagonal entries,

$$\varepsilon \gg |m_{\rm B} - m_{\rm A}|, \tag{13}$$

the eigenstates become approximately symmetric and antisymmetric combinations of $\nu_{\rm A}$ and $\nu_{\rm B}$. These combinations behave effectively as Majorana fermions, since each eigenstate is its own CPT conjugate up to a phase.

If, on the other hand, the diagonal entries dominate,

$$|m_{\rm B} - m_{\rm A}| \gg \varepsilon, \tag{14}$$

the eigenstates remain predominantly associated with a single sector. In this regime, the neutrinos behave effectively as Dirac fermions, with the two sectors playing the role of left- and right-handed components.

In RDCC, the relaxation-induced mixing is small but non-zero. This places the system in an intermediate regime where neither the pure Dirac nor the pure Majorana limit is realized exactly. Instead, the effective nature of the neutrinos depends on the projection onto sector A and the relative size of the diagonal and off-diagonal terms. The Dirac/Majorana distinction is thus not a fundamental property of the global state but an emergent feature of the thermodynamic arrow.

This perspective has important implications for neutrinoless double beta decay. Since the effective Majorana mass depends on the off-diagonal mixing term ε, the decay rate is naturally suppressed in RDCC. This suppression provides a distinctive observational signature that we explore in the next section.

7 Observational Consequences and Predictions

The relaxation-induced neutrino sector of RDCC leads to a number of distinctive observational signatures. These signatures arise from the interplay between the bipartite global state, the temperature asymmetry between the sectors, and the suppressed inter-sectoral mixing. In this section we summarize the most important predictions for cosmology and laboratory experiments.

7.1 Prediction 1: Correlation between $\Delta N_{\rm eff}$ and the Lightest Neutrino Mass

The temperature asymmetry between the sectors implies a mild excess in the effective number of relativistic species,

$$\Delta N_{\rm eff} \simeq 0.1\text{–}0.3. \tag{15}$$

Since the relaxation mechanism suppresses the masses more strongly for lighter flavors, the lightest neutrino mass is correlated with the temperature ratio $T_{\rm B}/T_{\rm A}$. This leads to a characteristic prediction:

$$m_{\rm lightest} \propto \left(\frac{T_{\rm B}}{T_{\rm A}}\right)^p, \tag{16}$$

where p is a positive exponent determined by the relaxation dynamics. As a result, a measurement of $\Delta N_{\rm eff}$ directly constrains the lightest neutrino mass within RDCC.

7.2 Prediction 2: Lower Bound on the Sum of Neutrino Masses

The relaxation-induced seesaw mechanism suppresses the lightest eigenvalue but leaves the heavier eigenvalues close to the sector-B mass scale. This leads to a lower bound on the sum of neutrino masses,

$$\Sigma m_\nu \gtrsim 0.06\,{\rm eV}, \tag{17}$$

consistent with the normal hierarchy. The inverted hierarchy requires a fine-tuned cancellation and is therefore disfavored. Future CMB and large-scale-structure surveys will be able to test this prediction.

7.3 Prediction 3: Suppressed Neutrinoless Double Beta Decay

The effective Majorana mass relevant for neutrinoless double beta decay is given by

$$m_{\beta\beta} \propto \varepsilon, \tag{18}$$

where ε is the relaxation-induced mixing term. Since ε is small, the decay rate is naturally suppressed in RDCC. This provides a distinctive signature: even if neutrinos have a Majorana component at the level of the global state, the observable decay rate may lie below the sensitivity of current and near-future experiments.

7.4 Prediction 4: Scale-Dependent Neutrino Free-Streaming

The sectoral mass splitting implies that the effective neutrino mass relevant for structure formation depends on the projection onto sector A. This leads to a mild scale dependence in the neutrino free-streaming length, which affects the matter power spectrum at intermediate scales. The effect is small but potentially detectable in high-precision surveys.

7.5 Prediction 5: Oscillation Signatures from Relaxation-Induced Mixing

The off-diagonal mixing term ε introduces a small correction to the standard oscillation probabilities. While the effect is too small to alter the leading oscillation parameters, it may produce subleading signatures in long-baseline experiments. These signatures include slight deviations from the expected energy dependence and small corrections to the effective mixing angles.

7.6 Prediction 6: Sectoral Mass Splitting as a Test of RDCC

The most distinctive prediction of RDCC is the sectoral mass splitting between $m_{\mathrm{A},i}$ and $m_{\mathrm{B},i}$. While the heavier sector is not directly observable, its imprint appears in the relaxation-induced suppression of the lightest eigenvalue. A precise measurement of the mass hierarchy, the sum of neutrino masses, and ΔN_{eff} would allow a consistency test of the RDCC neutrino sector.

Together, these predictions provide a coherent and testable framework for probing the neutrino sector within RDCC. The combination of cosmological and laboratory measurements offers a powerful avenue for testing the relaxation mechanism and the bipartite structure of the global state.

8 Conclusion

In this Companion we have developed the neutrino sector of the Relaxation-Driven Cyclic Cosmology (RDCC), completing one of the last major open components of the framework. Starting from the bipartite global state and the universal relaxation mechanism introduced in Companion X, we derived a relaxation-induced neutrino mass matrix that naturally generates suppressed masses, sectoral mass splittings, and an emergent Dirac/Majorana structure.

The key insight is that the neutrino sector is uniquely sensitive to the bipartite structure of the global state. The temperature asymmetry between the sectors leads to distinct diagonal mass contributions, while the relaxation dynamics induce a small but non-zero inter-sectoral mixing. Together, these effects produce a universal relaxation-induced seesaw mechanism that requires no new fields or high-scale physics.

This structure favors the normal mass hierarchy, predicts a lower bound on the sum of neutrino masses, and yields a correlation between ΔN_{eff} and the lightest neutrino mass. The emergent Dirac/Majorana nature of the neutrinos provides a natural explanation for the suppression of neutrinoless double beta decay, while the sectoral mass splitting leads to subtle but potentially observable signatures in cosmology and oscillation experiments.

The predictions summarized in this Companion offer a coherent and testable framework for probing the neutrino sector within RDCC. Future measurements of ΔN_{eff}, the mass hierarchy, the sum of neutrino masses, and the rate of neutrinoless double beta decay will provide powerful tests of the relaxation mechanism and the bipartite global state.

With the development of the neutrino sector, RDCC now offers a unified and internally consistent description of the thermal history, matter content, and dynamical evolution of the Universe. The framework continues to provide a rich set of observational signatures, making it a compelling alternative to conventional cosmological models.

A Diagonalization of the Relaxation-Induced Mass Matrix

For completeness we provide the explicit diagonalization of the 2×2 mass matrix,

$$M_\nu = \begin{pmatrix} m_{\text{A}} & \varepsilon \\ \varepsilon & m_{\text{B}} \end{pmatrix}. \tag{19}$$

The characteristic equation is

$$\lambda^2 - (m_{\text{A}} + m_{\text{B}})\lambda + (m_{\text{A}} m_{\text{B}} - \varepsilon^2) = 0, \tag{20}$$

with eigenvalues

$$\lambda_\pm = \frac{1}{2}\left[m_{\text{A}} + m_{\text{B}} \pm \sqrt{(m_{\text{B}} - m_{\text{A}})^2 + 4\varepsilon^2}\right]. \tag{21}$$

In the limit $\varepsilon \ll |m_{\text{B}} - m_{\text{A}}|$, the lighter eigenvalue becomes

$$m_{\text{light}} \simeq m_{\text{A}} - \frac{\varepsilon^2}{m_{\text{B}} - m_{\text{A}}}, \tag{22}$$

which defines the relaxation-induced seesaw.

B Temperature Asymmetry and Sectoral Mass Splitting

The temperature ratio between the sectors satisfies

$$\frac{T_{\text{B}}}{T_{\text{A}}} \simeq 0.5, \tag{23}$$

as derived in Companion X. The effective masses scale as

$$m_{\text{B}} = m_{\text{A}}\, f(T_{\text{B}}/T_{\text{A}}), \tag{24}$$

where f is a smooth function determined by the relaxation dynamics.

For $T_{\text{B}}/T_{\text{A}} < 1$, the function f generically yields $m_{\text{B}} > m_{\text{A}}$, leading to a sectoral mass splitting that is robust against variations in the microscopic parameters.

C Robustness of the Relaxation-Induced Seesaw

The suppression of the lightest eigenvalue depends only on the hierarchy

$$\varepsilon \ll |m_{\text{B}} - m_{\text{A}}|, \tag{25}$$

which is guaranteed by the smallness of the relaxation rate. The mechanism is therefore robust and does not require fine-tuning.

The seesaw remains stable under:

- moderate variations in the temperature ratio $T_{\mathrm{B}}/T_{\mathrm{A}}$,
- small perturbations in the diagonal entries m_{A} and m_{B},
- higher-order corrections to the relaxation rate.

This ensures that the predictions derived in the main text are insensitive to microscopic uncertainties and remain valid across the full parameter space of RDCC.

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